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What is a multiplying DAC?

JUNE 20, 2017 BY JANET HEATH

Multiplying digital-to-analog converters (MDACs) produce a (current) output signal that's a product of the given reference voltage and the code (i.e., a string of 0s and 1s) flowing through it. All data converters require a voltage reference (V_{REF}) and a typical, standard DAC needs a very stable fixed reference voltage in order to operate properly. However, the Multiplying Digital-to-Analog converter (MDAC) is a low-noise, precision DAC that uses a *varying* reference voltage. MDACs are often used in test and measurement equipment and other applications that require low noise. $V_{REF} \times I_{OUT} \times$

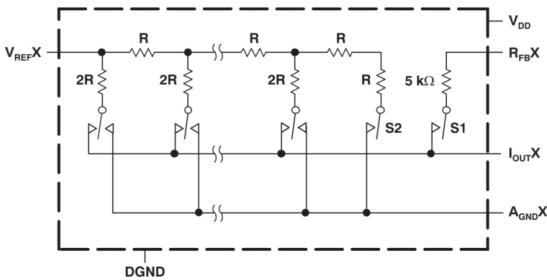


Figure 1. One equivalent MDAC channel of the TI DAC8812 (digital connections omitted for clarity), containing two current-steering R-2R ladder DACs. The RFBX pin is provided as a matching feedback resistor for connection to the inverting input of an external I-to-V op amp. (Source: DAC8812 Datasheet, Texas Instruments.)

Typically, analog devices like op amps and data converters have input, output, and reference voltages (for data converters) that are bounded by the "rails" set by the device's supply voltage (V_S). The MDAC is an exception, as it can use a V_{REF} that is much higher than the voltage supplied to the MDAC itself. The external V_{REF} , then, determines the full-scale output for an MDAC, not V_S . Another difference between MDACs and fixed-reference DACs is that MDACs produce output as a current signal rather than a voltage output. This is easily remedied with a **current-to-voltage converting op amp** following the MDAC to provide a voltage signal output.

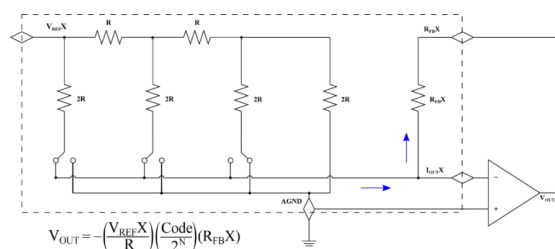


Figure 2. An MDAC with an I-to-V operational amplifier and matched, integrated feedback resistor. (Credit: K. C. Reese, III)

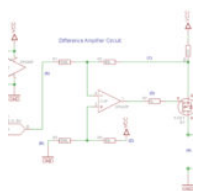
Another property of MDACs is that the reference impedance stays constant regardless of how the V_{REF} varies. That's right, the impedance of the MDAC measured at the V_{REF} pin stays constant regardless of the inputs to the MDAC or the value of the V_{REF} . The impedance of the MDAC output *does* vary with the code that is passing through the MDAC. The output current of the MDAC varies with the code flowing through the DAC. Therefore, looking at Figure 1, you can calculate the current at I_{OUTX} if you know the code traveling through the DAC and the equivalent series resistance in the DAC from V_{REF} using Kirchoff and Ohm's laws.

It's important to note that the output impedance of the MDAC is going to vary with the code-dependent current flowing out of the I_{OUT} pin. This poses potential issues for the flowing I-to-V converting op amp. The RFBX internal to the MDAC is matched to the impedances in the MDAC, which helps, but the I-to-V converting op amp should have low offset voltage to avoid introducing linearity errors, because offset voltage in the op amp will be affected by the

changing impedance it's getting at the I_{OUT} pin.

MDACs are used in several applications. For instance, if you have a digital design where you need to create an analog output with a fixed DC voltage as your resource. The MDAC can generate a waveform. An MDAC can also change the polarity and the amplitude of an AC signal. MDACs are known for precision (as long as they are paired with a precision or high-precision op amp). Major U.S. manufacturers of MDACs include Analog Devices and Texas Instruments.

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